

ENEL445 Project Planning Report: Optimisation Methods for Variational Inference

David Ewing (82171165)
ENEL445 – Applied Engineering Optimisation
University of Canterbury

March 17, 2026

1 Background and Problem Intuition

1.1 Why Variational Inference?

Bayesian inference provides principled uncertainty quantification but requires computing intractable posterior distributions. For most real-world models, the posterior integral $p(\boldsymbol{\theta} \mid \mathbf{y}) = p(\mathbf{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})/p(\mathbf{y})$ cannot be evaluated analytically. Variational Inference (VI) converts this inference problem into an **optimisation problem**, making it directly amenable to the gradient-based methods taught in ENEL445.

1.2 The Core Idea

Instead of computing the true posterior, VI finds the “best” approximation $q(\boldsymbol{\theta})$ from a tractable family by maximising the Evidence Lower Bound (ELBO). This transforms intractable integration into tractable optimisation—a natural fit for an optimisation course project.

2 Optimisation Problem Formulation

2.1 Decision Variables

For Bayesian linear regression with p predictors, the variational parameters (decision variables) are:

- $\boldsymbol{\mu}_\beta \in \mathbb{R}^p$: Posterior mean of regression coefficients
- $\boldsymbol{\Sigma}_\beta \in \mathbb{R}^{p \times p}$: Posterior covariance (symmetric positive definite)
- $a_e, b_e > 0$: Shape and rate parameters of Gamma distribution for precision

Total dimension: $d = p + \frac{p(p+1)}{2} + 2 = 11$ variables for $p = 3$ predictors.

2.2 Objective Function

Maximize the Evidence Lower Bound (ELBO):

$$\text{ELBO}(\boldsymbol{\phi}) = \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\beta}, \tau_e)] - \mathbb{E}_q[\log q(\boldsymbol{\beta}, \tau_e)] \quad (1)$$

where $\boldsymbol{\phi} = (\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, a_e, b_e)$ are the variational parameters.

2.3 Constraints

- $\Sigma_\beta \succ 0$: Covariance matrix must be positive definite
- $a_e, b_e > 0$: Gamma parameters must be positive
- Optional: Minimum variance constraints $\sigma_{\beta_j}^2 \geq \sigma_{\min}^2$ to prevent underdispersion

These constraints can be handled via parameterization (e.g., Cholesky decomposition for Σ_β) or penalty methods.

2.4 Computational Characteristics

- Gradient computation: $O(np^2)$ per evaluation (dominated by matrix operations)
- Hessian computation: $O(np^2 + d^3)$ for Newton's method
- Parameters exhibit coupling (changing μ_β affects optimal Σ_β)
- Must handle special functions: digamma $\psi(\cdot)$, trigamma $\psi'(\cdot)$ for Gamma distribution

3 Test Case: Bayesian Linear Regression

The test case uses Bayesian linear regression with:

$$\mathbf{y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \tau_e^{-1}\mathbf{I}_n) \quad (\text{likelihood}) \quad (2)$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda_\beta^{-1}\mathbf{I}_p) \quad (\text{prior on coefficients}) \quad (3)$$

$$\tau_e \sim \text{Gamma}(\alpha_e, \gamma_e) \quad (\text{prior on precision}) \quad (4)$$

Mean-field approximation:

$$q(\boldsymbol{\beta}, \tau_e) = q(\boldsymbol{\beta}) q(\tau_e) \quad (5)$$

$$q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}_\beta, \Sigma_\beta) \quad (6)$$

$$q(\tau_e) = \text{Gamma}(a_e, b_e) \quad (7)$$

Variational parameters: $\phi = (\boldsymbol{\mu}_\beta, \Sigma_\beta, a_e, b_e)$ with dimension $d = p + \frac{p(p+1)}{2} + 2$

For $p = 3$ predictors: $d = 3 + 6 + 2 = 11$ parameters to optimise.

4 Overview of Optimisation Methods

4.1 Baseline Method: CAVI

Coordinate Ascent Variational Inference (CAVI) is VI-specific and not a standard ENEL445 method. It updates each variational parameter block sequentially using closed-form solutions:

$$\Sigma_\beta \leftarrow \left(\frac{a_e}{b_e} \mathbf{X}^T \mathbf{X} \right)^{-1} \quad (8)$$

$$\mu_\beta \leftarrow \frac{a_e}{b_e} \Sigma_\beta \mathbf{X}^T \mathbf{y} \quad (9)$$

$$a_e \leftarrow \alpha_e + \frac{n}{2} \quad (10)$$

$$b_e \leftarrow \gamma_e + \frac{1}{2} [\|\mathbf{y} - \mathbf{X} \mu_\beta\|^2 + \text{tr}(\mathbf{X}^T \mathbf{X} \Sigma_\beta)] \quad (11)$$

Properties: Monotonic ELBO increase, linear convergence, $O(np^2)$ per iteration. Provides analytical benchmark for validation.

4.2 Course-Based Methods

The following methods from ENEL445 apply directly to maximizing ELBO

4.3 Mapping Course Methods to VI Problem

1. Gradient Ascent (Course: Pages 27-28, gradient descent)

- Standard iterative update: $\phi^{(t+1)} = \phi^{(t)} + \alpha_t \nabla \text{ELBO}(\phi^{(t)})$
- Line search from Chapter 3.2 (bisection/Wolfe conditions)
- Stopping criteria: $\|\nabla \text{ELBO}\| < \epsilon$
- Convergence: Linear rate

2. Newton's Method (Course: Equation 3.26)

- Standard Newton update: $\phi^{(t+1)} = \phi^{(t)} + \alpha_t [\nabla^2 \text{ELBO}]^{-1} \nabla \text{ELBO}$
- Requires computing and inverting the Hessian matrix
- Computational cost: $O(d^3)$ per iteration for Hessian inversion
- Convergence: Quadratic rate near optimum
- Challenge: Hessian may not be positive definite away from optimum

3. BFGS Quasi-Newton (Course: Equations 4.50-4.51, one of five required methods from Project 2)

- BFGS update formula builds inverse Hessian approximation
- Uses gradient differences: $\mathbf{s}_k = \phi^{(k)} - \phi^{(k-1)}$, $\mathbf{y}_k = \nabla \text{ELBO}(\phi^{(k)}) - \nabla \text{ELBO}(\phi^{(k-1)})$
- Woodbury identity for efficient updates of inverse Hessian
- Computational cost: $O(d^2)$ per iteration
- Convergence: Superlinear rate
- Widely regarded as best general-purpose optimiser for smooth unconstrained problems

4.4 Why These Methods?

- **Gradient Ascent:** Baseline first-order method, expected to be slowest but most robust
- **Newton’s Method:** Best convergence rate (quadratic) if Hessian well-conditioned
- **BFGS:** “Best general-purpose optimiser” for smooth problems (course recommendation)
- **CAVI:** Domain-specific baseline exploiting problem structure

Comparison will reveal whether general optimisation methods can compete with or outperform problem-specific algorithms

5 Solution Plan

5.1 Phase 1: Data Generation and Baseline (Completed)

Data simulation:

- Generate $n = 100$ observations, $p = 3$ predictors
- Design matrix $\mathbf{X} \sim \mathcal{N}(0, 1)$, true coefficients $\boldsymbol{\beta}_{\text{true}} = [1, -0.5, 2]^T$
- Noise: $\epsilon \sim \mathcal{N}(0, 0.5^2)$

Mathematical foundations (completed):

- Complete ELBO derivation for Bayesian linear regression
- CAVI closed-form update equations
- Gradient expressions: $\nabla_{\boldsymbol{\mu}_\beta} \text{ELBO}$, $\nabla_{\boldsymbol{\Sigma}_\beta} \text{ELBO}$, $\nabla_{a_e} \text{ELBO}$, $\nabla_{b_e} \text{ELBO}$
- Hessian block structure: diagonal and mixed blocks
- BFGS update formulation with Woodbury identity

All derivations documented in `vi_optimization_solution.tex` (13 pages)

5.2 Phase 2: Implementation (In Progress)

1. Data Generation (5%)

- Simulate data: $n = 100$ observations, $p = 3$ predictors
- Generate $\mathbf{X}$ from standard normal, $\boldsymbol{\beta}_{\text{true}} = [1, -0.5, 2]^T$
- Add noise: $\epsilon \sim \mathcal{N}(0, 0.5^2)$

2. Baseline Implementation: CAVI (15%)

- Implement closed-form coordinate updates
- Track ELBO convergence
- Stopping criterion: $|\text{ELBO}^{(t+1)} - \text{ELBO}^{(t)}| < 10^{-6}$

3. Method 1: Gradient Ascent (25%)

- Implement ELBO gradient computation
- Line search: bisection method or Wolfe conditions
- Stopping criterion: $\|\nabla \text{ELBO}\| < 10^{-6}$
- Track iteration count and ELBO values

4. **Method 2: Newton's Method** (25%)

- Implement Hessian computation (block structure)
- Regularization: add $\lambda \mathbf{I}$ if Hessian not positive definite
- Line search for step size α_t
- Track convergence and computational cost

5. **Method 3: BFGS Quasi-Newton** (25%)

- Initialise inverse Hessian approximation: $\mathbf{B}_0 = \mathbf{I}_d$
- Implement BFGS update formula with Woodbury identity
- Line search with Wolfe conditions
- Track superlinear convergence

6. **Comparison and Analysis** (5%)

- Compare convergence speed (iterations to 10^{-6} tolerance)
- Compare computational cost (time per iteration, total time)
- Compare solution quality (posterior approximation accuracy)
- Analyse when each method performs best

5.3 Success Metrics

1. **Convergence:** All methods reach ELBO optimum within tolerance
2. **Correctness:** Solutions match CAVI baseline (analytical benchmark)
3. **Performance hierarchy:**
 - Expected: BFGS > Newton > Gradient Ascent (in convergence speed)
 - CAVI likely fastest for this conjugate problem
4. **Understanding:** Clear explanation of why each method performs as it does

6 Progress To Date

6.1 Completed Work

1. **Problem formulation:** Full optimisation problem specified with decision variables, objective function (ELBO), and constraints
2. **Mathematical derivations:** 13-page technical document containing:
 - Complete ELBO derivation from first principles

- CAVI algorithm with closed-form updates
 - Analytical gradient expressions for all 11 variational parameters
 - Hessian block computations for Newton’s method
 - BFGS quasi-Newton formulation
 - Penalty method formulation for constrained optimisation
3. **Method selection justification:** Demonstrated direct mapping between course material (Chapters 3-4) and VI optimisation:
- Gradient ascent $\leftrightarrow$ Algorithm 3.2 (gradient descent)
 - Newton’s method $\leftrightarrow$ Equation 3.26
 - BFGS $\leftrightarrow$ Equations 4.50-4.51 (one of five required project methods)
4. **Validation strategy:** Established CAVI as analytical benchmark for correctness checking

6.2 Preliminary Results

While full implementation is pending, the mathematical analysis reveals:

- **Gradient structure:** $\nabla_{\boldsymbol{\mu}_\beta} \text{ELBO} \propto \mathbf{X}^T(\mathbf{y} - \mathbf{X}\boldsymbol{\mu}_\beta)$ resembles least-squares gradient, suggesting ELBO surface is well-behaved
- **Hessian properties:** Diagonal blocks are negative definite (ELBO is locally concave), supporting Newton’s method applicability
- **Dimensionality:** With $d = 11$ parameters, Hessian inversion costs $\approx 1,300$ flops—manageable for Newton’s method without L-BFGS reduction
- **Parameter coupling:** Mixed Hessian blocks non-zero, indicating gradient-based methods exploiting joint updates should outperform coordinate ascent

6.3 Next Steps

1. Implement CAVI baseline in Python (Week 7)
2. Implement gradient ascent with line search (Week 7)
3. Implement Newton’s method with regularization (Week 8)
4. Implement BFGS quasi-Newton (Week 8)
5. Monte Carlo comparison: convergence iterations, wall-clock time, ELBO values (Week 8-9)
6. Final report with performance analysis (Week 12)

7 Alignment with Course Requirements

This project directly addresses the core learning objectives of ENEL445:

1. **Optimisation formulation:** Convert inference problem to unconstrained optimisation
2. **Gradient computation:** Analytical derivation and numerical implementation
3. **Method implementation:** Code gradient ascent, Newton, and BFGS from scratch
4. **Line search:** Apply bisection or Wolfe conditions for step size selection
5. **Stopping criteria:** Implement gradient norm and function value convergence tests
6. **Performance comparison:** Empirical evaluation of convergence rates and computational costs
7. **Method selection:** Understand when to use each optimisation algorithm

Key insight: The VI problem is *not* a toy problem. It represents a real computational challenge in modern machine learning, making the comparison of optimisation methods both pedagogically valuable and practically relevant.

8 Timeline

March 17, 2026	Mathematical derivations completed
March 18, 2026	Data generation and CAVI implementation
March 19, 2026	Gradient ascent and Newton's method implementation
March 20, 2026	BFGS implementation and comparison analysis
March 20, 2026	Progress report submission

9 Summary

The Variational Inference optimisation problem provides an ideal testbed for implementing and comparing gradient-based methods from ENEL445. The problem:

- **Directly applies** course material (gradient descent, Newton, BFGS)
- **Requires** full implementation of algorithms with line search and stopping criteria
- **Enables** meaningful performance comparison on realistic problem
- **Demonstrates** practical application of optimization theory to modern machine learning

The mathematical foundation is complete. Implementation and empirical evaluation will demonstrate mastery of optimisation methods taught in ENEL445.